

Unit - 3(Cell Structure and Function) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



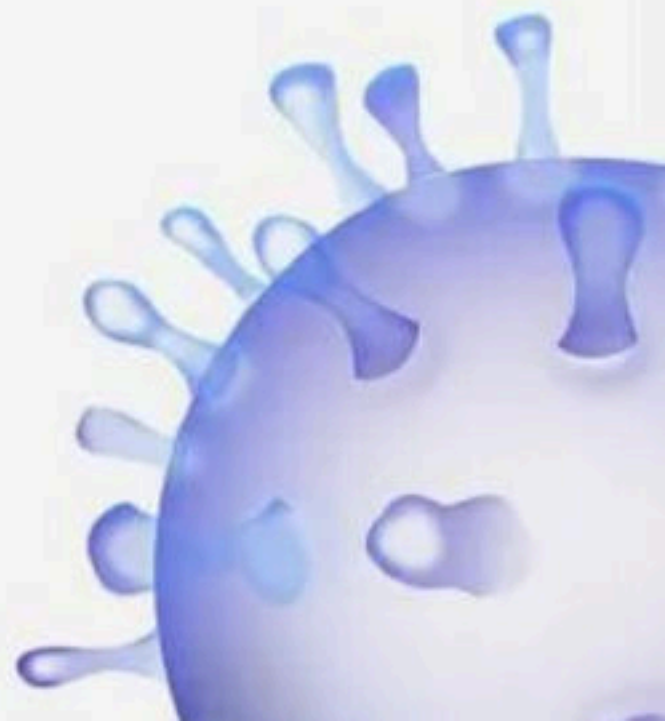
Good evening
↓
audible? ✓
Ready? ✓

Unit – 3 (Cell Structure and Function)

200 MCQs ✓



BIOLOGYLIVE



1. Centrioles are associated with :-

- (1) DNA synthesis
- (2) DNA replication
- ✓(3) Spindle formation ✓
- (4) Respiration



Title: Unit – 3 (Cell Structure and Function)

② **Assertion : Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles together form the endomembrane system.**

Reason : Functions of these organelles are coordinated so, they are considered together as a system.

- ✓ (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false



3. Unicellular organisms are capable of :-

- (1) Independent existence ✓
- (2) Performing essential functions of life ✓
- ✓ (3) Both (1) and (2)
- (4) None of these



Title: Unit – 3 (Cell Structure and Function)

4. Statement A : An elaborate network of filamentous proteinaceous structures present in the cytoplasm is called as cytoskeleton.

Statement B : Cytoskeleton consisting of microtubules, microfilaments and intermediate filaments.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both Statements are correct.

(4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

5. Statement A : The golgi apparatus principally performs the function of packaging materials.

Statement B : Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both Statements are correct.

(4) Both Statements are incorrect.



6. Assembly of 60S and 40S subunits of ribosome gives

(1) 50S

(2) 70S (50S + 30S)

✓ (3) 80S

(4) 100S



7. Plasma membrane is-

(1) Semipermeable

✓ (2) Selective / differentially permeable

(3) Permeable

(4) Non-permeable



⑧ The centrioles in a centrosome lie ⊥ to each other in which each has an organisation like the cartwheel.

(1) Longitudinal, cartwheel

(2) Parallel, cartwheel

✓ (3) Perpendicular, cartwheel

(4) Cylindrical, cartwheel



9) Which of the following membrane proteins lie on the surface of the cell?

↳ Peripheral

(1) Integral proteins

✓ (2) Peripheral proteins

(3) Both (1) and (2)

(4) Conjugate proteins



10. Which of the following have single membrane surrounding them?

- (1) Microbodies ✓
- (2) Vacuoles ✓
- (3) Golgi apparatus ✓
- ✓ (4) All of these



11. Plasmids are

- (1) Naked genomic DNA
- ✓ (2) Smaller DNA than genomic DNA
- (3) Enveloped DNA
- (4) Chromosomal DNA



12. Who was the first one to see a live cell?

☒ (1) Anton Von Leeuwenhoek

(2) Robert Leeuwenhoek

(3) Robert Brown

(4) None of these



13 Statement A : Ribosomes are composed of deoxy
ribonucleic acid and proteins. ✗

Statement B : Ribosomes are surrounded by ^{no} membrane. ✗

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

✓ (4) Both Statements are incorrect.



14. Identify the membrane that surrounds the vacuole?

(1) Protoplast

✓ (2) Tonoplast

(3) Cell membrane

(4) None of these



15. Small particles["] projecting from the inner membrane of mitochondria are known as :-

- (1) Microsome
- (2) Macrosome
- ✓ (3) F₀-F₁ particles ✓
- (4) Lysosomes



11.1mP

16. The stroma of chloroplast contains enzymes required for synthesis of

(1) Carbohydrates ✓

(2) Proteins ✓

(3) Fats

✓ (4) Both 1 and 2



17. Which component of plasma membrane increase the fluidity?

- (1) Phospholipids
- ✓ (2) Cholesterol
- (3) Carbohydrates
- (4) Proteins



18. Which one of the following is "genetically active" and contain loosely packed DNA?

(1) Heterochromatin

✓ (2) Euchromatin

(3) Nucleolus

(4) Mitochondria



19. Spindle fibres attach on to_____.

(1) Telomere of the chromosome

✓ (2) Kinetochore of the chromosome ✓

(3) Centromere of the chromosome

(4) Kinetosome of the chromosome



20. Choose the incorrect statement. :-

- (1) Human red blood cells are about $7.0\text{ }\mu\text{m}$ in diameter
- (2) Bacteria are $3\text{--}5\text{ }\mu\text{m}$ in length
- (3) The largest cell is the egg of an ostrich
- ☒ (4) Mycoplasma is the ~~largest~~ cell



Title: Unit – 3 (Cell Structure and Function)

21. Nuclear DNA exists as a complex of proteins called – **Chromatin**
that **condenses in to** – **during** – **cell division**

(1) Chromatids, Chromosomes, Cell division

(2) Chromosomes, Chromatin, Interphase

✓ (3) Chromatin, Chromosome, Cell Division

(4) Chromatin, Chromosome, Interphase



22. Non-membrane bound organelles present in animal cell that helps in cell division.

(1) Nucleus

(2) Ribosomes

✓✓ (3) Centriole → sphere → cytoplasm

(4) Golgi apparatus



Title: Unit – 3 (Cell Structure and Function)

23. Statement A : Singer and Nicolson in ¹⁹⁷²~~1927~~ proposed ~~x~~
widely accepted asfluid mosaic model of plasma membrane.

Statement B : According to this, the quasifluid nature of
lipid enables lateral movement of proteins within the overall bilayer. ✓

- (1) Statement A is correct but Statement B is incorrect.
- ✓(2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

24. Choose the right sequence/route of the secretory product?

(1) RER → GB → Lysosome → Nuclear membrane →

Plasma membrane

☒ (2) ER → Vesicles → Cis region of GB → Trans region of

GB → Plasma membrane

(3) ER → Vesicles → Trans region of GB → Cis region of

GB → Vesicles → Plasma membrane

(4) Lysosome → ER → GB → Vesicles → Cell membrane



25. Axoneme having 9+2 doublet microtubule arrangement
is found in :-

- (1) Centriole
- (2) Flagella
- (3) Cilia
- ✓ (4) Cilia and Flagella



26 Plants cell differs from animals cell in having -

- (1) Cell membrane, plastid and cell wall
- (2) Cell wall, plastid, centriole
- ✓ (3) Chloroplast, plastid and cell wall ✓
- ✓ (4) Large vacuole, plastid and cell wall ✓



Title: Unit – 3 (Cell Structure and Function)

27. Statement A : All living organisms are composed of cell or product of cells. ✓✓

Statement B : Unicellular organisms are composed of many cells. ✗

- ✓(1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

28. Select the correct statement:

V. Imp

(1) The chloroplast contains cristae and carotenoid pigments.

→ colourless X

(2) Leucoplast contains fat soluble carotenoid pigments like carotene, xanthophylls.

✓ (3) Plastid is easily observed under microscope.

(4) Chloroplast is a single membrane bound organelle.
Double



Title: Unit – 3 (Cell Structure and Function)

29. Statement A : Human red blood cells are about 7.0 μm in diameter.

Statement B : The shape of the cell may vary with the function they perform.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- ✓ (3) Both Statements are correct.
- (4) Both Statements are incorrect.



30. Statement A : Passive transport does not require energy. ✓✓

Statement B : Active transport also does not require ~~energy~~. ✗

✓(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.



31. Which one of the following has majority of the chloroplasts? ↓

- ✓ (1) Mesophyll cells of the leaves
- (2) Mesophyll cells of the stem
- (3) Mesophyll cells of the roots
- (4) Mesophyll cells of the flowers



32. Which cell organelle connects nuclear envelope with cell membrane?

(1) Nucleus itself

(2) Golgi body

✓ (3) Endoplasmic reticulum ✓

(4) Mitochondria



33. Cellular organelles with membranes are :-

- ✓ (1) Lysosomes, Golgi apparatus and mitochondria
- (2) Nuclei, ribosomes^x and endoplasmic reticulum
- (3) Chromosomes, ribosomes^x and endoplasmic reticulum
- (4) Endoplasmic reticulum, ribosome^x and nuclei



34. Statement A : Anton Von Leeuwenhoek first saw and described a live cell. ✓

Statement B : Robert Brown later discovered the nucleus nucleolus. ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



35. Microbodies are present in

(1) Plants

(2) Animals

✓ (3) Both 1 and 2 ✓

(4) None of the above



36. Which one of the following is formed by packaging in golgi apparatus?

- ☒ (1) Lysosomes
- (2) Endoplasmic reticulum
- (3) Both (1) and (2)
- (4) Mitochondria



37. Which of the following is a type of plastid?

- (1) Chloroplast
- (2) Chromoplast
- (3) Leucoplast
- ☒ (4) All of these



Title: Unit – 3 (Cell Structure and Function)

38. Statement A : In prokaryotes the genetic material is basically enveloped by a nuclear membrane. ✗

Statement B : Many bacteria have small circular DNA outside the genomic DNA, called Plasmids. ✓

(1) Statement-A is correct but Statement-B is incorrect.

✓(2) Statement-A is incorrect but Statement-B is correct.

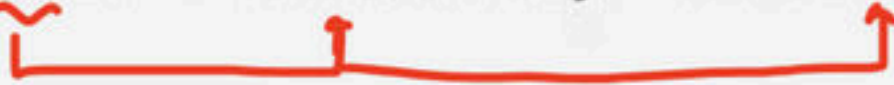
(3) Both Statements are correct.

(4) Both Statements are incorrect.



39. Which of the following sequence is correct?

- (1) Basal body → Cilium/flagellum → Centriole
- (2) Cilium/flagellum → Basal body → Centriole
- (3) Basal body → Centriole → Flagellum/ Cilium
- ✓ (4) Centriole → Basal body → Cilium/ Flagellum



40. What is the length of mycoplasma cell?

(1) 3 mm

✓ (2) 0.3 μm

(3) 3 μm

(4) 30 μm

↓
0.3 μm



Title: Unit – 3 (Cell Structure and Function)

V. Imf

41. Biochemical investigation revealed that the cell
membranes also possess

(1) Lipids and Carbohydrate

(2) Carbohydrate only

(3) Protein only

(4) Protein and Carbohydrate *along with lipid*

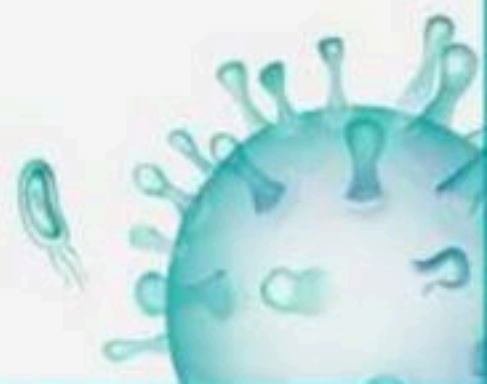


Title: Unit – 3 (Cell Structure and Function)

42. Statement A : In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm. ✓

Statement B : The cytoplasm is the main arena of cellular activities in only plant cells but not in animal cells. ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

43. Which of the following statements regarding cilia is correct?

(1) The axoneme possesses only one microtubul

✓ (2) The axoneme possesses a number of microtubules
running parallel to the long axis

(3) The axoneme possesses a number of microtubules
running perependicular to the long axis

(4) It varies depending on the organisms



44. Microtubules, motor proteins, and actin filaments are all part of the

(1) Mechanism of photosynthesis that occurs in chloroplasts.

(2) Rough ER in prokaryotic cells

✓(3) Cytoskeleton of eukaryotic cells. ✓

(4) Process that moves small molecules across cell membranes.



45. Chlorophyll pigments are present in :-

- ✓ (1) Thylakoid
- (2) Stroma
- (3) Outer membrane
- (4) Inner membrane



46. Bacterial flagellum has

(1) Filament – longest

(2) Hook

(3) Basal body

✓ (4) All of these



47 The ability of proteins to move within the membrane is measured as

(1) Plasticity

(2) Flexibility

✓ (3) Fluidity

(4) None of these



Title: Unit – 3 (Cell Structure and Function)

48. Statement A : Matthias Schleiden, a German ^{bot.}~~Zoologist~~ observed that all plants are composed of different kinds of cells X

Statement B : Theodore Schwann, a British ^{Zoal.}~~Botanist~~ X
studied different types of animal cells and
reported plasma membrane

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- ✓ (4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

49. Assertion : F₀-F₁ particles are present in the inner mitochondrial membrane. ✓

Reason : An electron gradient formed on the inner mitochondrial membrane forms

ATP. ✗ proton

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

✓ (3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false



Title: Unit – 3 (Cell Structure and Function)

50. What is true for a prokaryotic cell?

- i. Prokaryotic cell is usually 0.1-2.0 micron.
- ii. They are smaller in size
- iii. Greatly vary in shape and size
- iv. Multiply much rapidly than eukaryotes

(1) i, ii only

(2) i, ii, iii only

(3) i, iii, iv only

☒ (4) All of these



51. Chromatophores take part in :

- (1) Respiration
- ✓ (2) Photosynthesis
- (3) Growth
- (4) Movement



52 Nucleolus is :-

- (1) Rounded structure found in cytoplasm near nucleus.
- ☒ (2) Rounded structure inside nucleus and having rRNA. ✓
- (3) Rod-shaped structure in cytoplasm near the nucleus.
- (4) None of the above.



53. In flagella, there are radial spokes.

(1) 6

(2) 7

(3) 1

☒ (4) 9



54. Kinetochore is

(1) Centromere



✓(2) Protein discs at primary constriction ✓

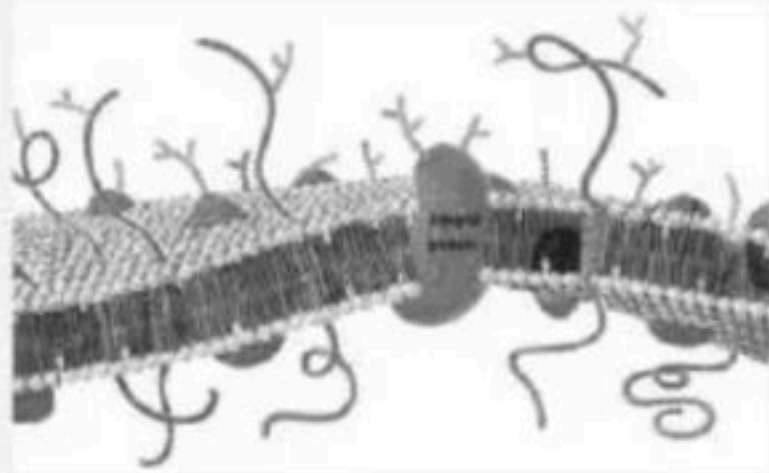
(3) Protein discs at secondary constriction

(4) DNA at centromere



Title: Unit – 3 (Cell Structure and Function)

55. Find out correct statement about the given diagram. A Scientists, B-Year, C- Structure.



(1) A - Singer and Nicolson, B - 1972, C – Nuclear membrane

✓ (2) A - Singer and Nicolson, B - 1972, C - cell membrane

(3) A - Robert brown, B - 1831, C - Nucleus

(4) A - Matthias Schleiden, B - 1838, C - Bacteria



56. Identify the most appropriate statement about the centrioles in a centrosome

a. Each of the peripheral fibril is a triplet. ✓

b. The adjacent triplets are linked. ✓

(1) a is correct

(2) b is correct

✓(3) Both are correct

(4) Both are incorrect



57. What are flat membranous tubules connecting the different grana called?

- (1) Thylakoid
- (2) Granum
- (3) Stroma
- ☒ (4) Stroma lamellae ☒



58. Which one of the following is ground substance of chloroplast?

- ✓ (1) Stroma
- (2) Granum
- (3) Thylakoid
- (4) Stroma lamella



59. Which among the following are small bristles like fibres sprouting out of the cell?

(1) Pili

☒ (2) Fimbriae

(3) Flagella

(4) Mitochondria



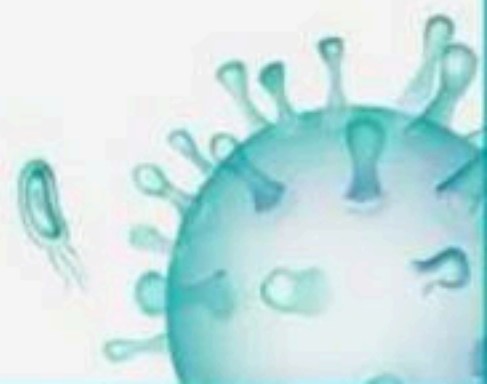
60. Lysosome are generally found in

- (1) Animal cells
- (2) Plant cells
- ✓ (3) both (1) and (2)
- (4) Bacterial cells



61. The function of cytoskeleton is

- (1) To facilitate intracellular transport ✓
- (2) To maintain cell shape and structure ✓
- (3) To provide support to the organelles ✓
- (4) All of the above ✓



62. The following statements tells us about a particular structure, identify it

i. It is ^{selectively} semi-permeable in nature and interacts with the outside world.

ii. It is similar structurally to that of the eukaryotes.

(1) Capsule

(2) Cell wall

☒ (3) Cell membrane

(4) Cytoplasm



63. A common characteristic feature of plant sieve tube cells and most mammalian erythrocytes is :-

- (1) Absence of nucleus and haemoglobin
- (2) Absence of nucleus and cell wall
- (3) Absence of mitochondria
- ✓ (4) Absence of nucleus



64. Assertion : Plastids are semiautonomous organelles.

Reason : Leucoplasts, Chromoplasts, Chloroplasts are plastids.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

☒ (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false



65. The axoneme usually has 1 pair of centrally located microtubules and 9 pairs of doublets of radially arranged peripheral microtubules,

(1) Nine, one

✓ (2) One, nine

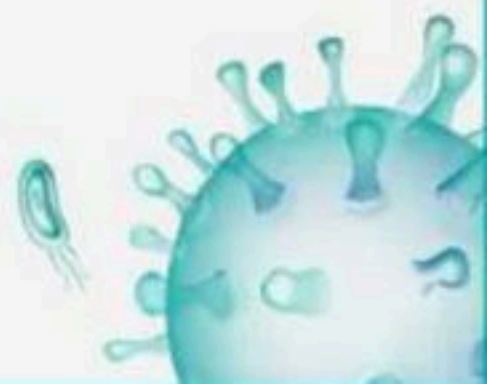
(3) Nine, zero

(4) Zero, nine



↓,
66 Actively functional nucleus shows-

- (1) Large nucleolus, diffused chromatin and no nuclear pores
- ✓ (2) Large nucleolus, diffused chromatin and more nuclear pores
- (3) Large nucleolus, compact chromatin and many pores
- (4) No nucleus, diffused chromatin and small nuclear pores



67. Golgi apparatus is often seen associated with :-

(1) Mitochondria

✓ (2) ER

(3) Lysosome

(4) Nucleus



68. Which of the following is incorrect matching?

✓ (1) Round and biconcave – ~~WBC~~ ^{RBC}

(2) Long and narrow – Columnar cell

(3) Elongated – Tracheid

(4) Long and branched – Nerve cells



69. The DNA present in chloroplast is:

- (1) Linear, double stranded
- ✓ (2) Circular, double stranded
- (3) Linear, single stranded
- (4) Circular, single stranded



70. Which of the following are not hydrolytic enzyme?

- (1) Nucleases
- (2) Proteases
- (3) Carbohydrases
- ✓ (4) Ligases



Title: Unit – 3 (Cell Structure and Function)

71. In which part of the nucleus does ribosome is
produced :-

- (1) Within nuclear pore
- (2) In the chromatin
- ✓ (3) In the area of the nucleolus
- (4) Ribosome production doesnot occur in the nuclei



Title: Unit – 3 (Cell Structure and Function)

72. In bacterial cilia, the arrangement of microtubules is

(1) $9 + 1$

(2) $9 + 0$

(3) $9 + 2 \rightarrow$ eukaryotes

✓ (4) 0



73. Lipids are arranged within the membrane with

(1) Polar heads towards innerside and the hydrophobic tails towards outside

(2) Both heads and tails towards outside

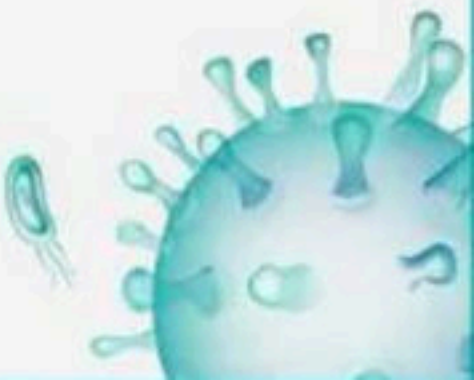
☒ (3) Polar heads towards outside and tails towards inside

(4) Both heads and tails towards innerside



74. Cell wall in plants consists of 202 S

- (1) Lignin, hemicellulose, pectin and lipid
- ✓ (2) Cellulose, hemicellulose, pectins and proteins.
- (3) Lignin, hemicelluloses, protein and lipid
- (4) Hemicellulose, cellulose, tubulin and lignin



15 Select the wrong statement:-

- (1) Bacteria exhibits 4 basic shapes
- ☒ (2) Pili and fimbriae are mainly involved in motility of bacterial cells
- (3) Cyanobacteria lack flagellated cells
- (4) Mycoplasma is a wall less microorganism



76. The _____ of the cell directs its overall activities as well as houses its genetic material .

(1) Nucleolus

(2) Endoplasmic reticulum

✓ (3) Nucleus

(4) Centromere



77 Which one of the following is absent in animal cell?

(1) Cellwall only



✓ (2) Plastids and cell wall

(3) Plastids only

(4) Cytoplasm



Title: Unit – 3 (Cell Structure and Function)

78. Statement A : Chloroplast contains ^{Small} ^{Stranded} double circular DNA molecule and ribosomes. X

Statement B : The ribosomes of the chloroplasts are ^{Smaller} larger than the cytoplasmic ribosomes.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) ☒ Both Statements are incorrect.



79. Which of the following is not a feature of the plasmids?

- (1) Independent replication
- (2) Circular structure
- (3) Transferable
- ☒ (4) Single stranded ☒



Title: Unit – 3 (Cell Structure and Function)

80. A few cells and associated entities are listed. Which of them represents the correct ascending order of the size relative to each other

(1) Mitochondrion < Paramecium < Human < erythrocyte < E. coli

(2) Virus < Protein < Paramecium < Chloroplast

✓ (3) Protein < Virus < Mitochondrion < Chloroplast < human RBC

(4) Nucleus < Protein < Paramecium < Chloroplast



81. The colour of leucoplast is

- (1) Yellow
- (2) Green
- (3) Orange
- (4) Colourless



82. Bacterial cell envelope is composed of :

- (1) Outermost cell wall followed by glycocalyx and plasma membrane
- (2) Outermost plasma membrane followed by cell wall and glycocalyx
- ✓ (3) Outermost glycocalyx followed by cell wall and plasma membrane
- (4) Plasma membrane and glycocalyx only



83. Of the two membranes of the chloroplast, the inner chloroplast membrane is _____

202 S

- (1) Totally impermeable
- (2) Relatively more permeable
- ✓ (3) Relatively less permeable.
- (4) None are correct



Title: Unit – 3 (Cell Structure and Function)

84. What is the ratio of microtubules in centrioles and flagella? $\rightarrow 9 \times 2 = 18$

$$\downarrow \\ (9 \times 3) = 27$$

$$\underline{27:18}$$

✓ (1) 27:18

(2) 20:27

(3) 1:1

(4) 20:9



Title: Unit – 3 (Cell Structure and Function)

85. Statement A : Chromosome with two equal arms is called submetacentric chromosome.

Statement B : Chromosome with one shorter arm and one longer arm is called ^{sub-}metacentric chromosome.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) ☒ Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

86 ER divides the ^{intracellular} space into ² distinct compartments

(1) Intracellular, three

(2) Extracellular, two

✓ (3) Intracellular, two

(4) Extracellular, three



87. Which of the following organelle is present only in animal cells?

(1) Membrane -bound - Mitochondria

✓ (2) Non-membrane -bound - Centriole ✓

(3) Membrane -bound - Vacuole

(4) Membrane -bound - Centriole



88. Centrosome is an organelle usually containing

- ✓ (1) Two cylindrical structures called centrioles ↪
- (2) Two spherical structures called centrioles.
- (3) One cylindrical structure called centriole
- (4) One spherical structures called centrioles.



Title: Unit – 3 (Cell Structure and Function)

89. Statement A : Na⁺/K⁺ Pump is an example of Passive ^{active} transporter. X

Statement B : Movement of alcohol ^{water} by diffusion is called osmosis. X

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- ✓ (4) Both Statements are incorrect.



90. Select the correct answer.

A. Prokaryotes have something unique in the form of inclusions ✓

B. Inclusions are essentially infoldings of cell membrane ✗

✓ (1) A is correct

(2) B is correct

(3) Both are correct

(4) Both are incorrect



Title: Unit – 3 (Cell Structure and Function)

91. Select the wrong statements from the following:

(1) Both chloroplast and mitochondria contain an inner and outer membrane

✓ (2) Both chloroplast and mitochondria have an internal compartment, always called as stroma X Chloropl.

(3) The chloroplasts are generally much larger than mitochondria ✓

(4) Both chloroplast and mitochondria contain DNA and 70 S ribosomes ✓



Title: Unit – 3 (Cell Structure and Function)

92. Statement A : In Amoeba the contractile vacuole is important for osmoregulation and excretion. ✓

2025

Statement B : In many cells, as in protists, contractile vacuoles are formed by engulfing the food particles.
food
x

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



93. Select the correct answer :-

(A) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins ✓

(B) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates only ✗

- ✓ (1) A is correct only
- (2) B is correct only
- (3) Both are correct
- (4) None are correct



94. Lipid bilayer is

(1) Hydrophilic

(2) Hydrophobic

✓ (3) Hydrophilic and Hydrophobic ✓

(4) Depends on the surrounding medium



Title: Unit – 3 (Cell Structure and Function)

95. Statement A : The ^{Pili}~~fimbriae~~ are elongated tubular structures made of a special protein. X

Statement B : The pili are small bristle like fibres sprouting out of the cell. X

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- ✓ (4) Both Statements are incorrect.



96 Cis and trans face of golgi body are ____ and ____
respectively.

- ✓ (1) Convex-forming, Concave -maturing
- (2) Concave-maturing, Convex-forming
- (3) Convex-maturing, Convex -forming
- (4) Concave-forming-, Concave maturing



97. A cell, which is very active in the synthesis and secretion of proteins, would be expected to have-

(1) Equal amount of RER and SER

(2) More SER than RER

✓ (3) More RER than SER

(4) It is always variable



98. For the study of structure of nucleus, the best cell is

- (1) Cell in any stage of mitosis
- (2) Cell in the late prophase ✗
- (3) Cell in the sphase
- ✓ (4) Cell in the interphase



99. What is the size of a typical eukaryotic cell?

(1) 1-2 μm

✓ (2) 10-20 μm

(3) 10-20 mm

(4) 10-20 cm



100. Identify the round and oval plant cell and amoeboid human cell respectively?

(1) Mesophyll cells, RBCs

✓ (2) Mesophyll cells, WBCs

(3) Columnar cells, Mesophyll cells

(4) Mesophyll cells, Nerve cell



101. Which of the following is correct about human Haemoglobin (Hb)?

- (1) Made up to 2-alpha and 2-beta subunits
- (2) Present in erythrocytes ✓
- (3) Use to carry O_2 and CO_2 ✓
- ✓ (4) All of these



102. The percentage of carbon in earths crust is

(1) 65%

(2) 46.6%

(3) 18.5%

✓ (4) 0.03%



103. Which of the following is an example of Phospholipid?

(1) Prostaglandins

✓ (2) Lecithin

(3) Bile acids

(4) Glycerol



104. With reference to enzymes, which one of the following statement is true?

(1) Apoenzyme = Holoenzyme + Coenzyme

✓ (2) Holoenzyme = Apoenzyme + Coenzyme

(3) Coenzyme = Apoenzyme + Holoenzyme

(4) Holoenzyme = Coenzyme - Apoenzyme



Title: Unit – 3 (Cell Structure and Function)

105. Statement A : When ice melting to water is an example of physical chemical reaction. ✗

Statement B : Hydrolysis of starch into glucose is a physical chemical process. ✗

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both Statements are correct.

✓ (4) Both Statements are incorrect.



106. An amino acid under certain conditions has both positive and negative charges simultaneously in the same molecule. Such a form of amino acid is called

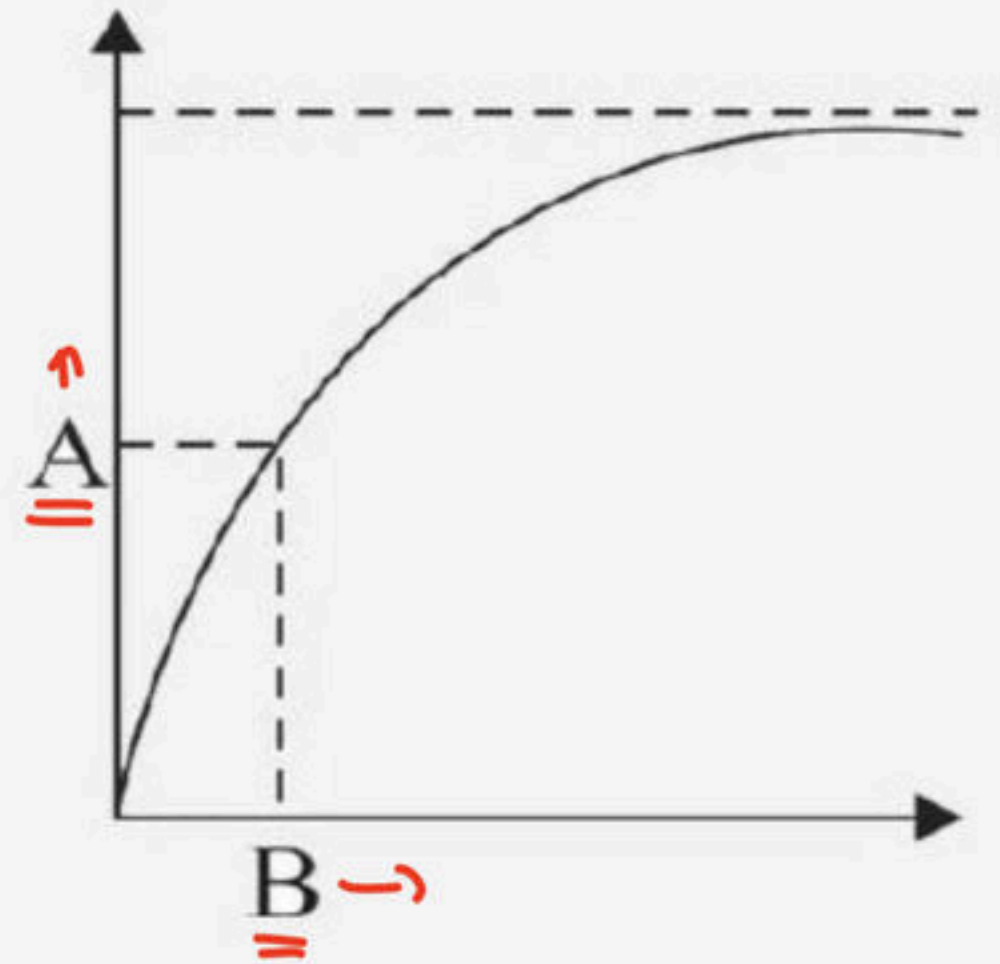
- (1) Acidic form
- (2) Basic form
- (3) Aromatic form
- ✓ (4) Zwitter ionic form ✓



Title: Unit – 3 (Cell Structure and Function)

107. Identify A and B in the given diagram?

- (1) A - Velocity of reaction, B - Concentration of substrate
- (2) A - Velocity of reaction, B - Temperature
- ✓(3) A - Velocity of reaction, B - K_m
- (4) A - Temperature, B - Velocity of reaction



108. Triglycerides consist of a glycerol molecule bonded to
how many numbers of fatty acids?

(1) 2

☒ (2) 3

(3) 4

(4) 1



Title: Unit – 3 (Cell Structure and Function)

109. ✓ Assertion : A coenzyme or metal ion tightly bound to enzyme protein is called prosthetic group.

✗ Reason : A complete catalytically active enzyme together with its bound prosthetic group is called apoenzyme ✗

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

✓ (3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.



Title: Unit – 3 (Cell Structure and Function)

110. Similarity in DNA and RNA is that

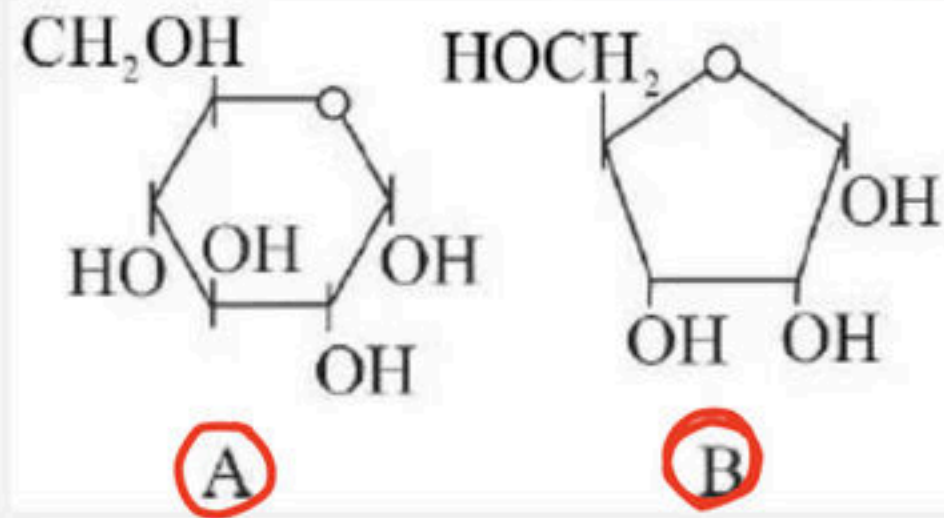
V. Imp

- (1) Both are polymers of nucleotides
- (2) Both have similar pyrimidines
- (3) Both have similar sugar
- (4) Both are non genetic material in man



Title: Unit – 3 (Cell Structure and Function)

111. Identify A and B in the given diagram?



anmolsharma12385@gmail.com

(1) A - Fructose, B - Glucose

✓ (2) A - Glucose, B - Ribose ✓

(3) A - Ribose, B - Glucose

(4) A - Galactose, B - Ribose



112. A non-proteinaceous enzyme is :-

- (1) Ligase
- (2) Deoxyribonuclease
- (3) Lysozyme
- ☒ (4) Ribozyme



113. An acid soluble compound formed by phosphorylation of nucleoside is called :-

- (1) Nitrogen base
- (2) Adenine
- (3) Sugar phosphate
- ✓ (4) Nucleotide ✓



114. Flow of metabolites through metabolic pathway has ____

- (1) Indefinite rate and direction
- ☒ (2) Definite rate and direction
- (3) Definite rate and no direction
- (4) No fix rate but direction



115. Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

2023

- I. Substrate enzyme complex formation. ②
- II. Free enzyme ready to bind with another substrate. ⑤
- III. Release of products. ④
- IV. Chemical bonds of the substrate broken. ③
- V. Substrate binding to active site. ①

Choose the correct answer from the options given below :

- | | |
|-----------------------------|---------------------------|
| ✓ (1) V - I - IV - III - II | (2) I - V - II - IV - III |
| (3) II - I - III - IV - V | (4) V - IV - III - II - I |



116. Nucleic acids exhibit a wide variety of _____ structures.

- ✓ (1) Secondary ↗
- (2) Primary
- (3) Tertiary
- (4) Quarternary



17 Cellulose consists of which type of unit?

(1) Galactose

✓ (2) Glucose

(3) Mannose

(4) Ribose



Title: Unit – 3 (Cell Structure and Function)

118. A homopolymer has only one type of building block called monomer repeated 'n' number of times. A heteropolymer has more than one type of monomer.

Proteins are heteropolymers usually made of

- ☒ (1) 20 types of monomers
- (2) 40 types of monomers
- (3) 30 types of monomers
- (4) only one type of monomer



119. Identify the mismatch

(1) Alkaloids – Morphine, Codeine

✓ (2) Lectins – ~~Abrin and Ricin~~

(3) Pigments – Carotenoids, Anthocyanin

(4) Terpenoides – Monoterpenes, Diterpenes



120. Select the correct matching for component percentage in living cell:

(1) Water – 7 to 59 % 70-90%

(2) Proteins – 1 to 15% 10-15%

(3) Carbohydrate – 12% 3%

(4) Nucleic Acid – 5 to 7% ✓



Title: Unit – 3 (Cell Structure and Function)

121. Assertion : Coenzyme is a non-protein group without which certain enzymes are inactive or incomplete.

Reason : Coenzymes not only provide a point of attachment of the chemical group being transformed, but also influence the properties of the group.

- ✓ (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



122. Half V_{max} is indicated by

(1) K_M

✓ (2) K_m

(3) k_M

(4) k_m

K_m



123. DNA and RNA are

- (1) Polypeptides
- ✓ (2) Polynucleotides
- (3) Polysaccharides
- (4) None of them



Title: Unit – 3 (Cell Structure and Function)

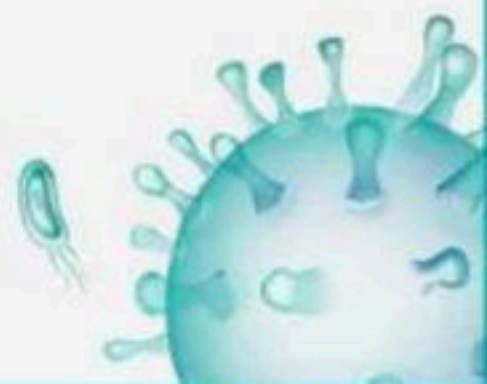
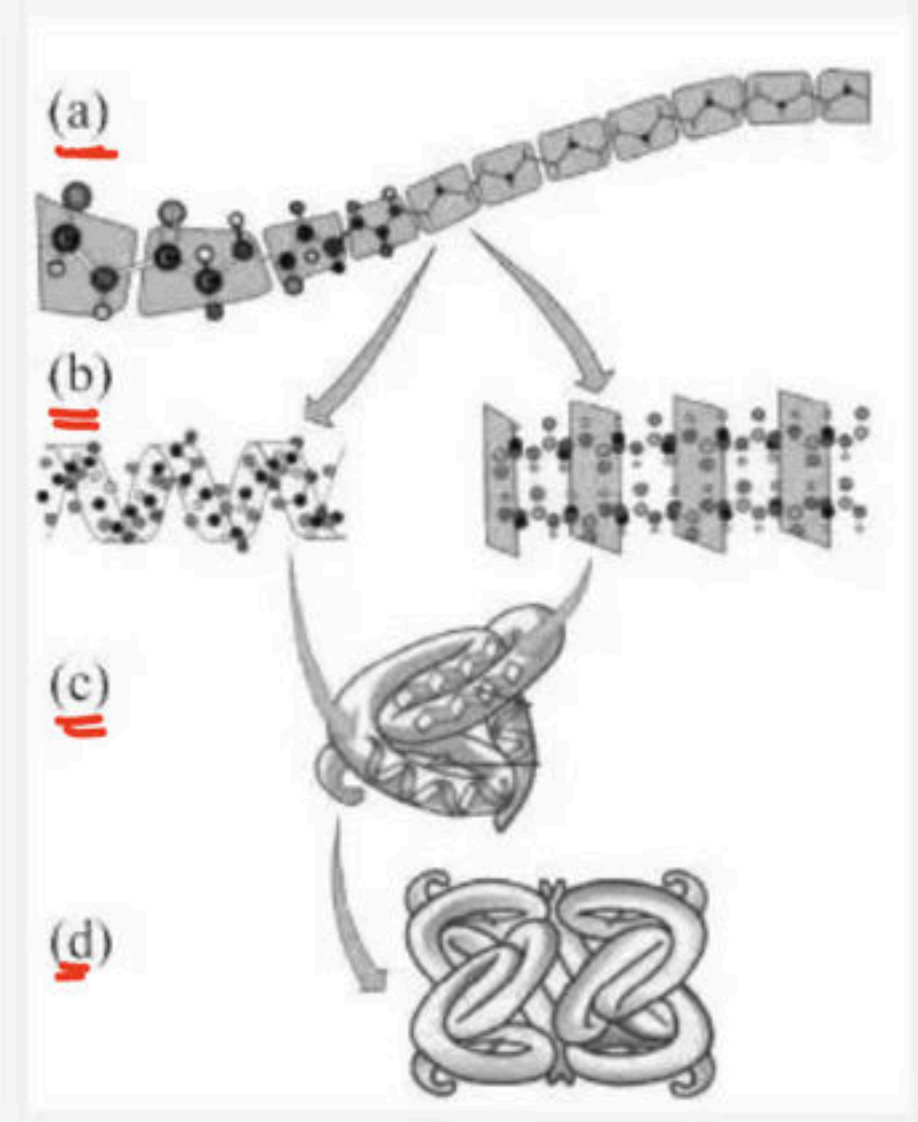
124. Following diagram shows different levels of protein structure. Identify the labels a to d in the diagram?

(1) a - Quaternary, b - Tertiary, c - Primary, d - Secondary

(2) a - Secondary, b - Primary, c - Quaternary, d - Tertiary

✓ (3) a - Primary, b - Secondary, c - Tertiary, d - Quaternary

(4) a - Secondary, b - Quaternary, c - Tertiary, d - Primary



125. Enzymes get denatured?

- ✓ (1) By destroying tertiary structure
- (2) By destroying primary structure
- (3) By destroying secondary structure
- (4) All of them



126. Enzymes catalysing the linking together of 2 compounds, e.g., enzymes which catalyse joining of CO, C-S, C-N, P-O etc. bonds

(1) Hydrolases

(2) Lyase

✓ (3) Ligase ✓

(4) Both (2) and (3)



127. The type of Polysaccharide present in a cotton fibre

- ☒ (1) Cellulose
- (2) Starch
- (3) Glycogen
- (4) Insulin



128. Which pyrimidine base is found in RNA only?

(1) Thymine

✓ (2) Uracil

(3) Cytosine

(4) 2 and 3



1700 → pen

129. Select the option which is not correct :-

(1) Malonate is a competitive inhibitor of succinic dehydrogenase

(2) Addition of [↓] lot of succinate does not reverse the inhibition of succinic dehydrogenase by malonate

(3) Increase in substrate concentration, increases the velocity of enzymatic reaction at first then it reaches maximum and further increment does not occur.

(4) All are correct



130. Which of the following is not a drug?

(1) Morphine

(2) Codeine

☒ (3) Carotenoids and anthocyanin ✓

(4) Both 1 and 2



131. An organic substance bound to an enzyme and essential for its activity is called

- (1) Isoenzyme
- ✓ (2) Coenzyme
- (3) Holoenzyme
- (4) Apoenzyme



132 Refer to the given reactions.

I. Adenine + X + Y → Adenylic acid

Choose the correct option for X and Y.

- | | |
|---|---------------------|
| (1) X - Phosphate group, | Y - Sugar molecule |
| (2) X - Nitrogen base, | Y - Sugar molecule |
| (3) X - Sugar molecule, | Y - Nitrogen base |
| ☒ (4) X - Sugar molecule, | Y - Phosphate group |



133. Which one is most abundant protein in the animal world?

(1) Trypsin

(2) Haemoglobin

✓ (3) Collagen

(4) Insulin



134. Which of the following is an example of sterol in animal tissue?

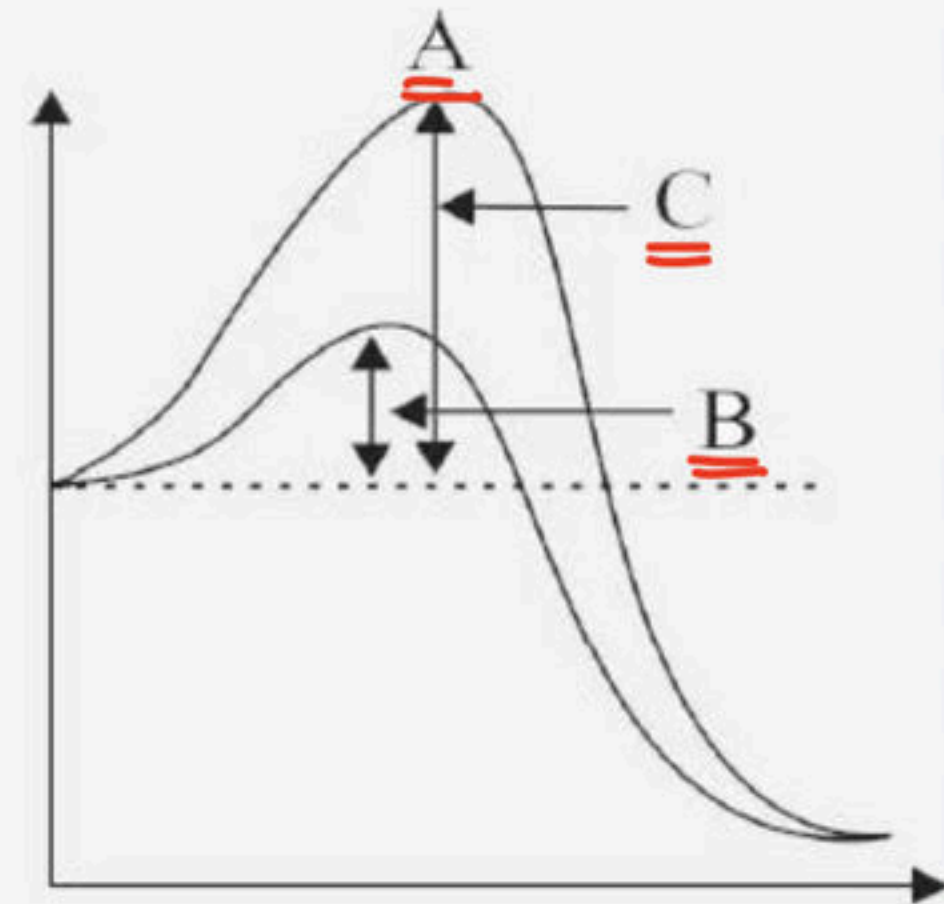
- ✓ (1) Cholesterol
- (2) Campesterol
- (3) Stigmasterol
- (4) More than one options are correct



Title: Unit – 3 (Cell Structure and Function)

135 Identify A, B and C in the given diagram?

- (1) A - Activation energy without enzyme, B - Transition state, C - Activation energy with enzyme
- (2) A - Transition state, B - Activation energy without enzyme, C - Potential energy.
- (3) A - Activation energy with enzyme, B - Transition state, C - Activation energy without enzyme
- ☒ (4) A - Transition state, B - Activation energy with enzyme, C - Activation energy without enzyme.



136. Assertion : Amino acids are monomers of nucleic acids ✗

Reason : Protein amino acids have an unlimited variety. ✗

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

✓ (4) Both Assertion and Reason are false.



137. Iodine test can detect the presence of

- ✓ (1) Starch
- (2) Cellulose
- (3) Both (1) and (2)
- (4) Chitin



138. What is/are the functions of proteins?

- (1) Transport nutrients across cell membrane
- (2) Fights infectious organisms
- (3) Acts as respiratory substrates
- ☒ (4) More than one options are correct



139. Select the incorrect statement from the following:

- (1) Starch forms helical secondary structures.
- ☒ (2) Starch does not give iodine test. ✓
- (3) Cellulose is a homopolysaccharide.
- (4) None of these



140. The inhibitor, which binds to the enzyme at site other than the active site and does not resemble the substrate in structure is called

- (1) Activator
- (2) Substrate analogue
- (3) Competitive inhibitor
- (4) ☒ Non-competitive inhibitor



Title: Unit – 3 (Cell Structure and Function)

141. Which among the following are the characteristic features of lipids?

- a. Lipids are important for cell membrane structure ✓
- b. They also functions as chemical messengers in human body ✓
- c. Lipids are also found in the blood cells and in the brain ✓
- d. Lipids are composed of carbon, hydrogen and oxygen in common and sometimes also contain phosphorous, sulphur, etc ✓

(1) a,c

(2) a,c,d

~~(3)~~ a,b,c,d

(4) b,c,d



142. How many types of cofactor can be identified?

(1) 1

(2) 2

✓ (3) 3

(4) 4



143. Hydrolysis of sucrose yields :

- ✓✓ (1) Glucose and fructose ←
- (2) Glucose and maltose
- (3) Glucose and galactose
- (4) Only glucose



144. Match the columns I (Component) with II (% of the total cellular mass) and choose the correct option.

Column I

A Nucleic acids

B Carbohydrates

C Proteins

D Lipids

Column II

P 2

Q 5-7

R 3

S 10-15

(1) A-R, B-S, C-P, D-Q

(3) A-Q, B-P, C-R, D-S

☒ (2) A-Q, B-R, C-S, D-P

(4) A-S, B-P, C-Q, D-R



Title: Unit – 3 (Cell Structure and Function)

145. The ¹¹catalytic efficiency ⁴of two different enzymes can be compared by the :-

(1) The K_m value ✗

✓(2) The optimum value

(3) Molecular size of enzyme

(4) The temperature optimum value



146. In the primary structure of protein

- (1) Left end represents → 1st amino acid (C-terminal amino acid)
- (2) Right end represents → Last amino acid (N terminal amino acid)
- ✓ (3) Left end represents → 1st amino acid (N-terminal amino acid)
- (4) Right end represents → 1st amino acid (C-terminal amino acid)



Title: Unit – 3 (Cell Structure and Function)

147. Statement A : Arachidonic acid has 20 carbon atoms
including
excluding the carboxyl carbon ✗

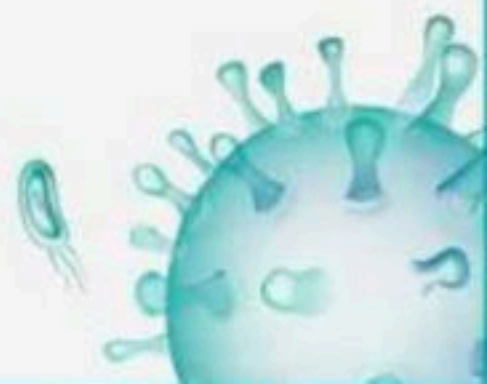
Statement B : Arachidonic acid is a saturated fatty acid.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- ✓ (4) Both Statements are incorrect.



148. Destructive effect of high temperature on enzyme action is due to

- (1) Inactivation of protein
- ✓ (2) Denaturation of protein
- (3) Acceleration of reverse reaction
- (4) Formation of protein



149. Select the correct option, which represents the homopolysaccharides made up of glucose monomers.

- (1) Sucrose, lactose, chitin
- ✓(2) Starch, glycogen, cellulose
- (3) Starch, inulin, peptidoglycan
- (4) Glucose, Starch, Cellulose



150. For its activity, carboxypeptidase requires:

- ✓ (1) Zinc
- (2) Iron
- (3) Niacin
- (4) Copper



151. Enzymes are divided into 6 classes each with 4-13 subclasses and named accordingly by a 4 digit number

(1) 4, 6, 13

(2) 6, 4, 13

✓ (3) 6, 4-13, 4

(4) Both (1) and (3)



152. Match the stages of meiosis in Column I to their characteristic features in Column II and select the correct option using the codes given below.

Column I		Column II	
A	Pachytene	1	Pairing of homologous chromosomes
B	Metaphase I	2	Terminalisation of chiasmata
C	Diakinesis	3	Crossing over takes place
D	Zygotene	4	Chomosomes align at equatorial plate

✓ (1) A-3, B-4, C-2, D-1

(2) A-4, B-1, C-2, D-3

(3) A-4, B-3, C-1, D-2

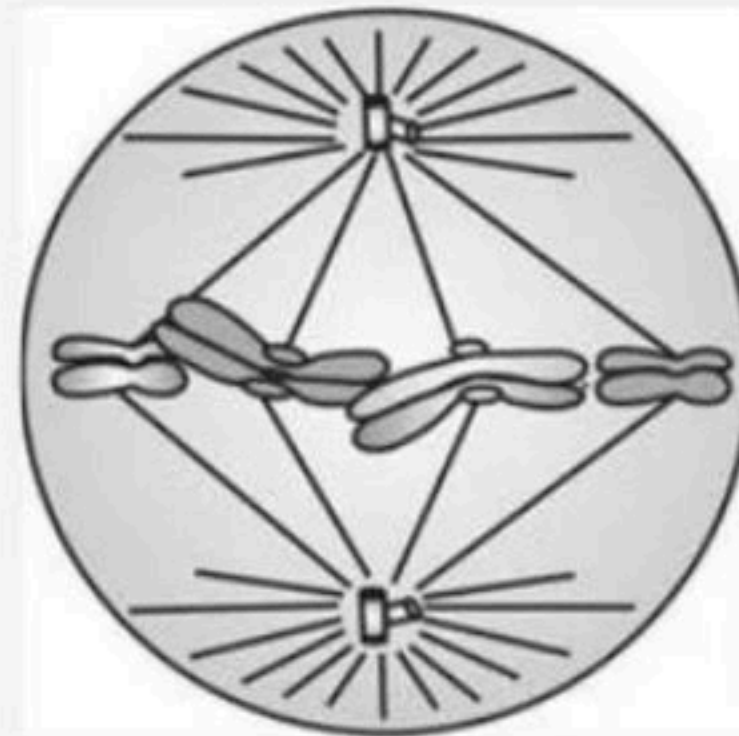
(4) A-4, B-2, C-1, D-3



Title: Unit – 3 (Cell Structure and Function)

153. Given diagram represents metaphase of mitosis, chromosomes are arranged at the equator which are made up of -

- ✓ (1) Two sister chromatids
- (2) Two non-sister chromatids
- (3) Two homologous chromosomes
- (4) Two non-homologous chromosomes



Title: Unit – 3 (Cell Structure and Function)

154. Statement A : Prophase which is the first stage of Cytokinesis of mitosis follows the ~~G₁~~ and ~~G₂~~ phases of interphase.

Statement B : Prophase is marked by the completion of condensation of chromosomal material.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



155. Statement A : A typical eukaryotic cell divide once in approximately every 22 hours. ✗

Statement B : Yeast for example, can progress through the cell cycle in only about 30 minutes. ✗

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- ✓ (4) Both Statements are incorrect.



156. Which does not occurs in prophase:-

- (1) Formation of chromatin $\checkmark \rightarrow$ telophase
- (2) Movement of centrioles
- (3) Appearance of sister chromatids
- (4) Compaction of chromosome



157. Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres.

(1) Metaphase I

(2) Metaphase II

✓ (3) Anaphase I ✓

(4) Anaphase II



158. As compared to meiosis, in mitosis :-

- (1) Homologous chromosomes pair
- (2) Homologous chromosomes separate
- (3) Anaphase stage is absent
- ✓ (4) Prophase is shorter ✓



159. Match the columns I and II and choose the correct option.

Column I

A Synaptonemal complex formation

B Appears as tetrads

C Chromosomes become gradually visible under the light microscope

D X-shaped structures

Column II

P Pachytene

Q Leptotene

R Zygotene

S Chiasmata

☒ (1) A-R, B-P, C-Q, D-S

(2) A-Q, B-P, C-R, D-S

(3) A-S, B-P, C-R, D-Q

(4) A-P, B-Q, C-R, D-S



160. The spindle fibre attaches to the

(1) Centromere

(2) Centrosome

(3) Centriole

(4) Kinetochore



161. How many times does the DNA replicates in one cell cycle?

- ☒ (1) Once ✓
- ☐ (2) Twice
- ☐ (3) Many times
- ☐ (4) Not at all



162. In meiosis, the chromosome number

- ☒ (1) Is reduced by half
- ☐ (2) Increases twice
- ☐ (3) Sometimes increases, sometimes decreases
- ☐ (4) None of the above



Title: Unit – 3 (Cell Structure and Function)

163. Which of the following statement(s) is/are true?

- (1) Cell plate represents the middle lamella between the walls of two adjacent cells. ✓
- (2) At the time of cytokinesis, organelles like mitochondria and plastids get distributed between the daughter cells. ✓
- (3) Cytokinesis in plant cell is centrifugal and takes place by cell-plate formation while animal cells by furrowing/cleavage and is centripetal. ✓
- (4) All of the above ✓



164. During gamete formation, the enzyme recombinase participates during

(1) Metaphase I

(2) Anaphase II

☒ (3) Prophase I ✓

(4) Prophase II



165. Which of the following types of cell cycle is known as equational division?

(1) Amitosis

(2) Meiosis

✓ (3) Mitosis ✓

(4) 2 and 3



Title: Unit – 3 (Cell Structure and Function)

166. Statement A : Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms.

Statement B : Fertilisation restores the diploid phase in the life cycle of sexually reproducing organisms.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- ☒ (3) Both Statements are correct. ✓
- (4) Both Statements are incorrect.



167 The phase of mitosis in which the centriole duplicates in the cytoplasm?

- ✓ (1) S-phase
- (2) G₁-phase
- (3) G₂-phase
- (4) None of these

60-70 per cent



Title: Unit – 3 (Cell Structure and Function)

168. Which of the following statement(event) is/ are true for mitotic telophase?

- (1) Nucleolus, GB and ER reform ✓
- (2) Nuclear envelope develops around the chromosome clusters ✓
- (3) Chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements. ✓
- (4) All of the above ✓



Title: Unit – 3 (Cell Structure and Function)

169. Match Column – I with Column - II.

Column I		Column II	
A	S phase	P	Proteins are synthesized
B	G_2 phase	Q	Inactive phase
C	Quiescent stage	R	Interval between mitosis and initiation of DNA replication
D	G_1 phase	S	DNA replication

(1) A - R, B - Q, C - P, D - S

(2) A - S, B - Q, C - R, D - P

✓ (3) A - S, B - P, C - Q, D - R

(4) A - Q, B - S, C - R, D - P



Title: Unit – 3 (Cell Structure and Function)

170. Assertion : Mitotic division is an equational division.

Reason : Second meiotic division is homotypic.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

☒ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



171. Synapsis occurs between :-

- (1) Spindle fibres and centromeres.
- (2) RNA and ribosomes.
- (3) A male and female gamete.
- ☒ (4) Two homologous chromosomes.



172. Which of the following statement is incorrect?

- (1) Prophase II is simpler than prophase I.
- (2) In leptotene stage the chromosomes become gradually visible under light microscope.
- (3) Nuclear membrane reappears only in telophase II not in telophase I. ✗
- (4) Anaphase II is characterized by the splitting of centromere.



173. In which stage of mitosis, the chromosomes are composed of two chromatids :-

- ☒ (1) Prophase and metaphase
- (2) Metaphase and telophase
- (3) Prophase and telophase
- (4) Prophase and anaphase



174. Which of the following is the longest phase of meiosis?

(1) Prophase II

(2) Metaphase I

☒ (3) Prophase I ✓

(4) Metaphase II



175. Statement A : Crossing over is the exchange of genetic material between two non-homologous chromosomes. ✗

Statement B : Crossing over is an enzyme-mediated process and the enzyme involved is called recombinase. ✓

(1) Statement-A is correct but Statement-B is incorrect.

✓ (2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.



176. The appearance of recombination nodules on homologous chromosomes during meiosis

characterizes:

(1) Bivalent

✓ (2) Sites at which crossing over occurs

(3) Terminalization

(4) Synaptonemal complex



177. What is the function of the enzyme 'recombinase' during meiosis?

- (1) Formation of synaptonemal complex
- ☒ (2) Crossing over between non-sister chromatids ✓
- (3) Condensation of chromosomes
- (4) Alignment of bivalent chromosomes on equatorial plate



178. Tetrad is made of

- (1) Four homologous chromosomes, each with one chromatids
- (2) Four non-homologous chromosomes
- (3) Four homologous chromosomes with four chromatids
- (4) ✓ Two homologous chromosomes, each with two chromatids ✓



Title: Unit – 3 (Cell Structure and Function)

179. Name the following stages of meiosis_

(1) A - Early prophase, B - Late prophase,

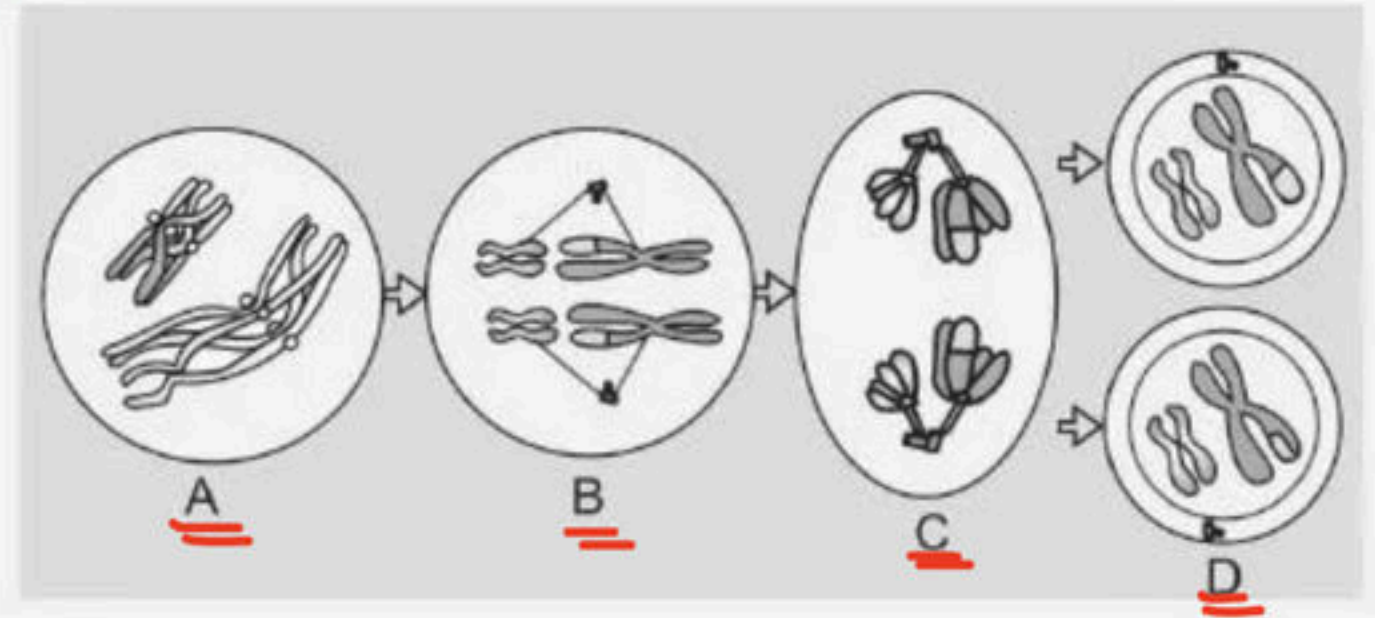
C - Anaphase, D - Telophase

(2) A - Prophase I, B - Metaphase I,

C - Anaphase II, D - Telophase II

(3) A - Prophase II, B - Metaphase II, C - Anaphase II, D - Telophase II

(4) A - Prophase I, B - Metaphase I, C - Anaphase I, D - Telophase I



180. The complex formed by a pair of synapsed homologous chromosomes is called _____

- ☒ (1) Bivalent
- (2) Axoneme
- (3) Equatorial plate
- (4) Kinetochore



Title: Unit – 3 (Cell Structure and Function)

181. Statement A : The beginning of diplotene is recognised by the dissolution of the synaptonemal complex ✓

Statement B : In oocytes of ~~all~~ some vertebrates, diplotene can last for months or years. X

- ✓(1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



182. Recombination nodule formation is a characteristic of which of the following phases?

(1) Zygotene

(2) Diplotene

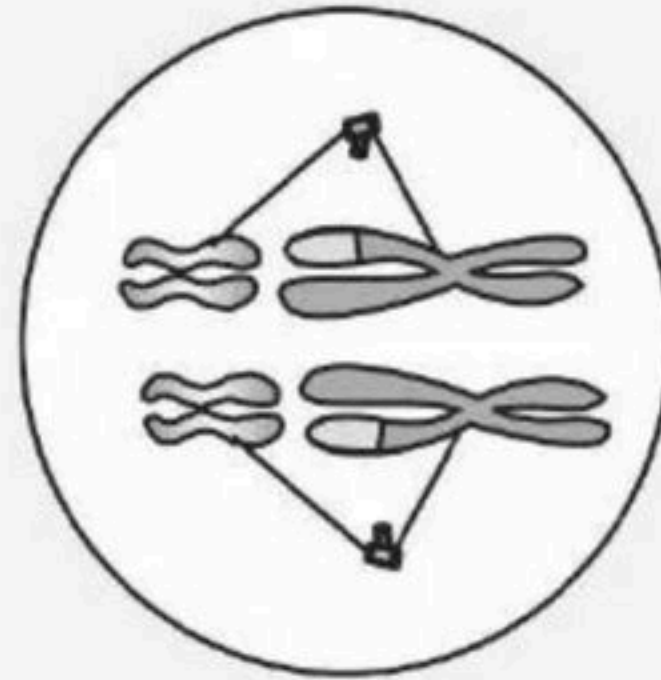
(3) Leptotene

(4) Pachytene



Title: Unit – 3 (Cell Structure and Function)

183. Following diagram represents metaphse I, in which
_____ are arranged at equator.



- ✓ (1) Bivalent chromosomes
- (2) Tetravalent chromosomes
- (3) Monovalent chromosomes
- (4) Both 1 and 2



184. The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

(1) Zygotene

✓ (2) Pachytene

(3) Diplotene

(4) Diakinesis



185. A bivalent of meiosis-I consists of

- (1) Two chromatids and one centromere
- (2) Two chromatids and two centromere
- ✓✓ (3) Four chromatids and two centromere
- (4) Four chromatids and four centromere



Title: Unit – 3 (Cell Structure and Function)

186. Statement A : During anaphase-I, both homologous chromosomes and sister chromatids separate.

Statement B : During Telophase-I, the nuclear membrane and nucleolus reappear.

(1) Statement-A is correct but Statement-B is incorrect.

✓(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.



▲ 3 • Asked by Binay Kuma...

Please help me with this doubt



Title: Unit – 3 (Cell Structure and Function)

187. In the phase shown by given diagram _

- (1) The homologous chromosome separates
- (2) Bivalent chromosome separates
- (3) Chromosomes start pairing
- ✓ (4) Cytokinesis follows and this is called dyad of cells



188. The X-shaped structures observed during diplotene are



✓ (1) Chiasmata

(2) Synaptonemal complex

(3) Bivalent complex

(4) 1 and 2



189. In diploid organisms, phenomenon of crossing over is responsible for

- (1) Linkages between genes
- ☒ (2) Recombination and variation ☒
- (3) Segregation between genes
- (4) Assortment of gene



190. Synaptonemal complex involves

- ✓ (1) Chromosome pairing
- (2) Chromosome movement
- (3) Chromosome segregation
- (4) Chromosome structure



191. The beginning of diplotene is characterized by

- (1) Recombination nodule formation
- (2) Synapsis
- ☒ (3) Dissolution of synaptonemal complex
- (4) Crossing over



Title: Unit – 3 (Cell Structure and Function)

192. Select the correct statements.

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I. Tetrad formation is seen during Leptotene. X

II. During Anaphase, the centromeres split and chromatids separate. ✓

III. Terminalization takes place during Pachytene. X

IV. Nucleolus, Golgi complex and ER are reformed during Telophase. ✓

V. Crossing over takes place between sister chromatids of homologous chromosome. X

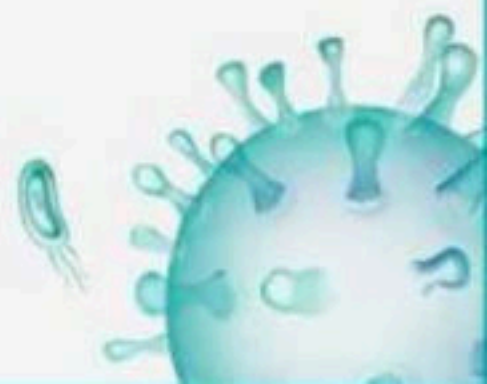
Choose the correct answer from the options given below:

(1) I and III only

✓ (2) II and IV only

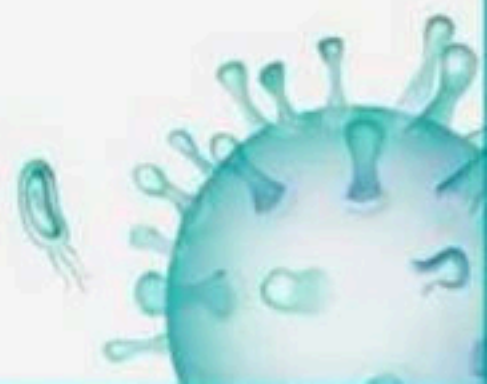
(3) I, III and V only

(4) II and V only



193. Which of the following represents the correct order in prophase I?

- (1) Zygotene, diplotene, pachytene, leptotene, diakinesis
- ✓ (2) Leptotene, zygotene, pachytene, diplotene, diakinesis
- (3) Leptotene, diplotene, pachytene, zygotene diakinesis
- (4) Pachytene, leptotene, zygotene, diplotene, diakinesis



194. In which stage of meiosis- I, the chromosome is thin, long and thread-like?

(1) Pachytene

(2) Zygotene

☒ (3) Leptotene ✓

(4) Diakinesis



195. The stage during which separation of the paired homologous chromosomes begins is

(1) Diakinesis

✓ (2) Diplotene ✓

(3) Pachytene

(4) Zygotene



196. Statement A : During leptotene stage the chromosomes become gradually visible under the light microscope. ✓

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Statement B : During diakinesis stage chromosomes start pairing together and this process of association is called synapsis. ✗

- (1) ✓ Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



197. Cross-like configuration when non-sister chromatids of a bivalent comes in contact during the meiosis-I are

✓ (1) Chiasmata

(2) Bivalents

(3) Synapsis

(4) 1 and 3



Title: Unit – 3 (Cell Structure and Function)

198. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Pairing of chromosomes together	P	Bivalent
B	Pair of synapsed homologous chromosomes	Q	Pachytene
C	Appearance of recombination nodules	R	Synapsis
D	Crossing over	S	Recombinase

(1) A-R, B-S, C-Q, D-P

(2) A-R, B-S, C-P, D-Q

(3) A-P, B-Q, C-S, D-Q

(4) A-R, B-P, C-Q, D-S



Title: Unit – 3 (Cell Structure and Function)

199. Statement A : During metaphase-I, the bivalent chromosomes align on the equatorial plate. ✓

Statement B : During anaphase-I, the microtubules from the opposite poles of the spindle attach to the kinetochore of homologous chromosomes. ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



Title: Unit – 3 (Cell Structure and Function)

100. Statement A : Prophase of the first meiotic division is typically longer and more complex when compared to prophase of mitosis ✓

Statement B : It has been further sub divided into the four sub phases, i.e leptotene, zygotene, pachytene and diplotene. ✗ *diakinesis*

✓ (1) Statement-A is correct but Statement-B is incorrect.

(2) Statement-A is incorrect but Statement-B is correct.

(3) Both Statements are correct.

(4) Both Statements are incorrect.



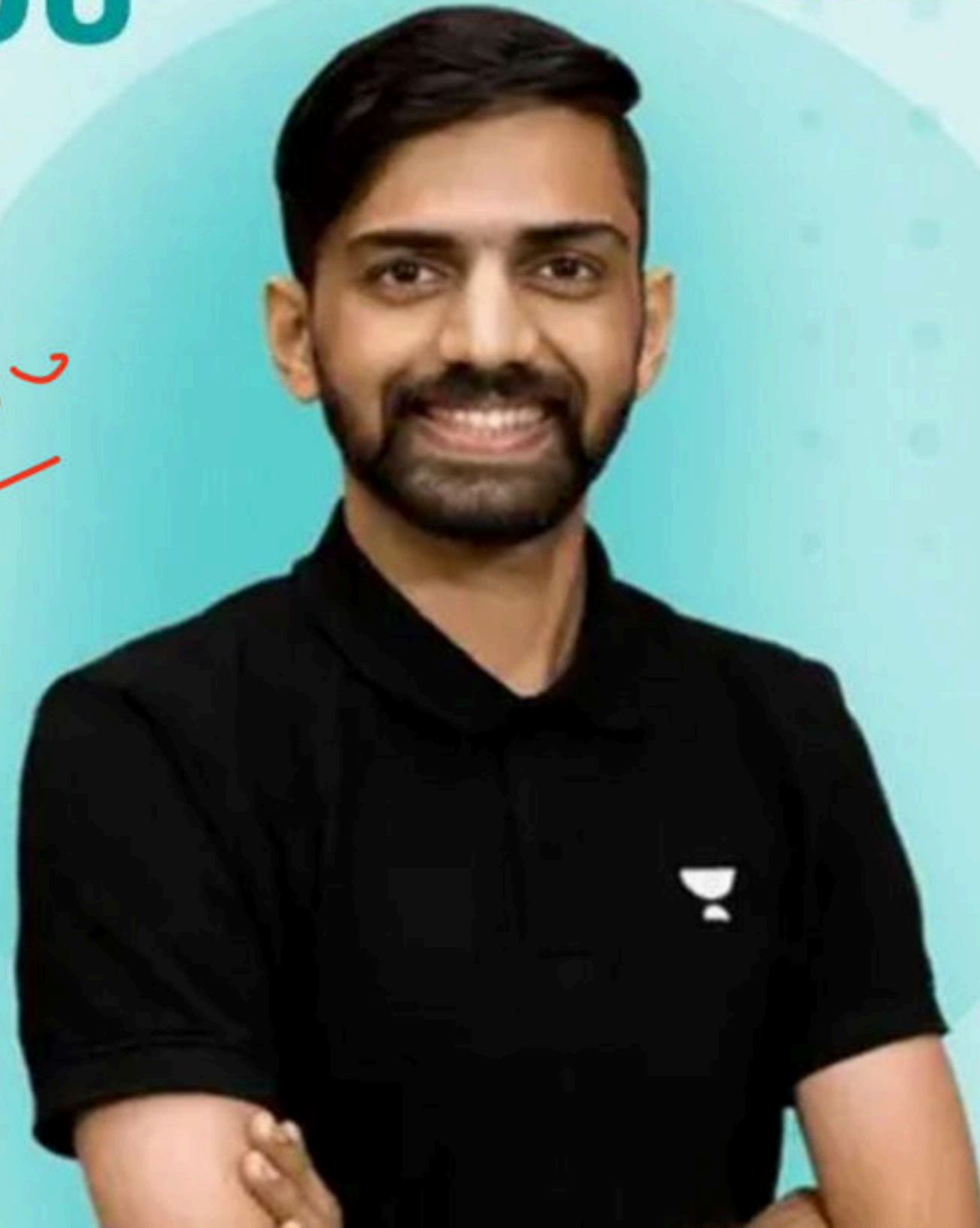
THANK YOU

@anmolsharmabiology



BIOLOGYLIVE

Thank You



6PM = $\Rightarrow$
plant physiol